Orbit-Evidence

Relational Validity Checks for Learning-Assisted Satellite Communication Experiments

A chronological split constrains order inside one dataset.
Some deployment-validity conditions are not in that dataset.

Anonymous Submission

Ordering alone is not deployment validity

Learning components now sit inside satellite communication software — correcting a propagated orbit, adapting a link parameter, gating a learned branch.

These systems are temporal, so we evaluate with a chronological split. That is *necessary*, and well established.

The claim of this paper

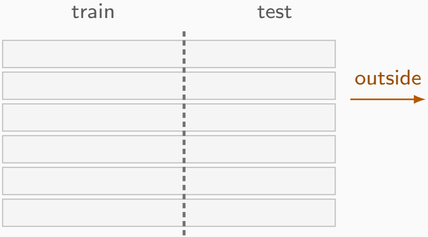
It is not sufficient — and the gap is one of *kind*, not degree.

A passing split can coexist with a deployment claim that was never checkable from the data in hand.

Train on the past, test on the future — and still not be able to say the experiment was valid.

Four obligations a split cannot establish

one realised dataset



a chronological split is a predicate over these rows

Availability
past-dated but unpublished

Row membership
which rows exist at all

Hidden state
what is not a column

Statistical unit
is the unit exchangeable

Each of the four has a counterexample that is not in the table the split examines.

The pivot: some validity is relational

across executions

run A ⟷ run B

same manifest, different output
⇒ the manifest is incomplete

across source states

source at decision ⟷ source later

present in the later source,
past-dated, yet unavailable
at the decision time

across aggregation levels

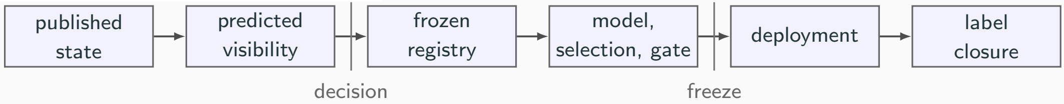
declared unit ⟷ coarser level

dependence at the coarser level
⇒ the *declared* unit is too fine

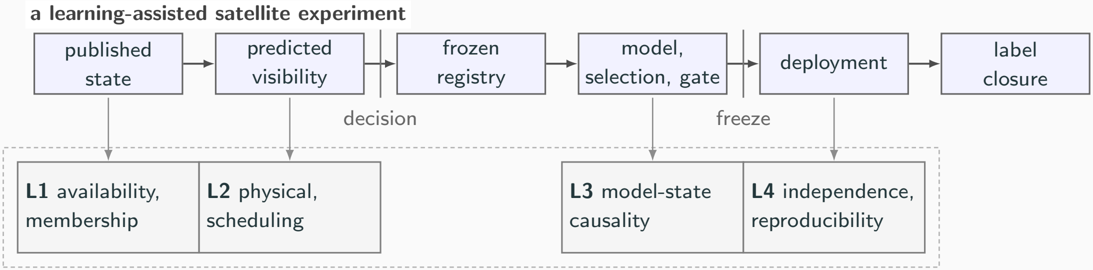
Row-local *validity is decidable from one realised run.* **Relational** *validity needs a second execution, a second source state, or a second aggregation level.*

Orbit-Evidence: validity as an executable contract

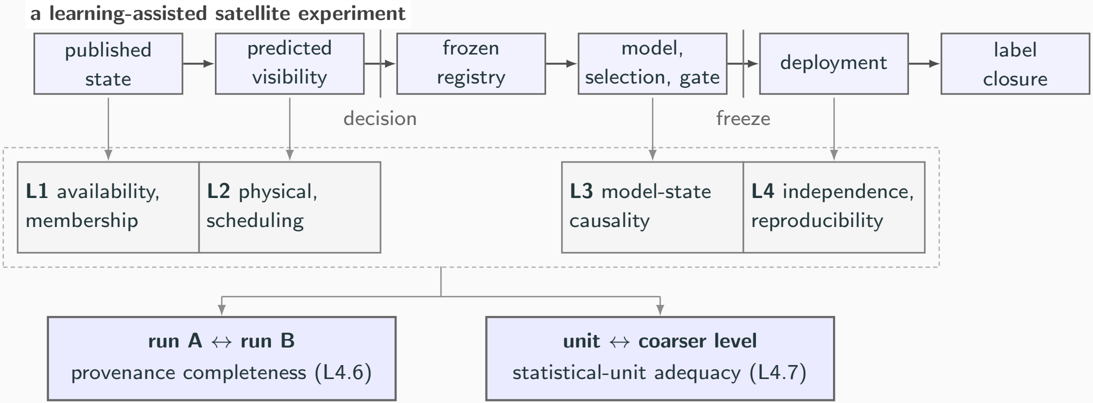
a learning-assisted satellite experiment



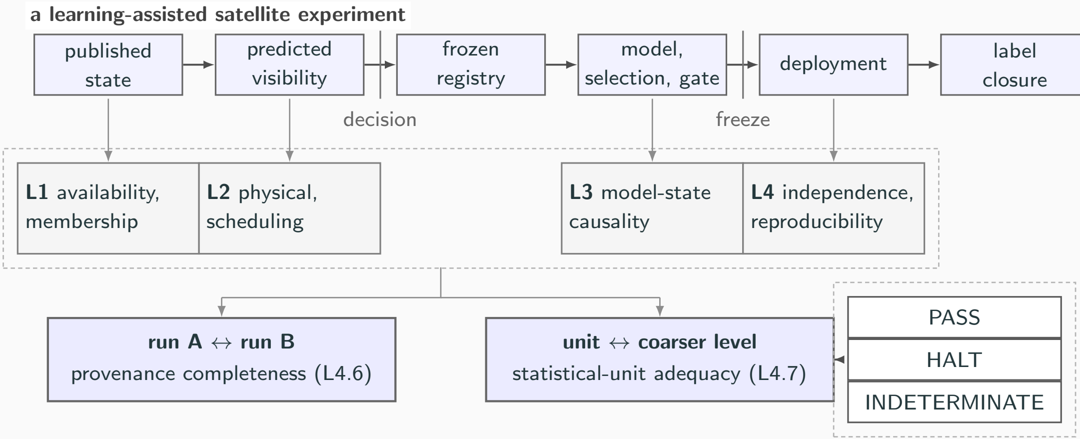
Orbit-Evidence: validity as an executable contract



Orbit-Evidence: validity as an executable contract



Orbit-Evidence: validity as an executable contract



PASS = not rejected HALT = obligation violated
INDETERMINATE = the design cannot decide

The two relational checks

L4.6 — differential provenance

Two runs, identical manifest hash, different output $\Rightarrow$ the manifest is incomplete.

Refutes a claim of completeness *without enumerating* the dependencies a manifest might omit. One-sided by construction: failure is evidence, success is not certification.

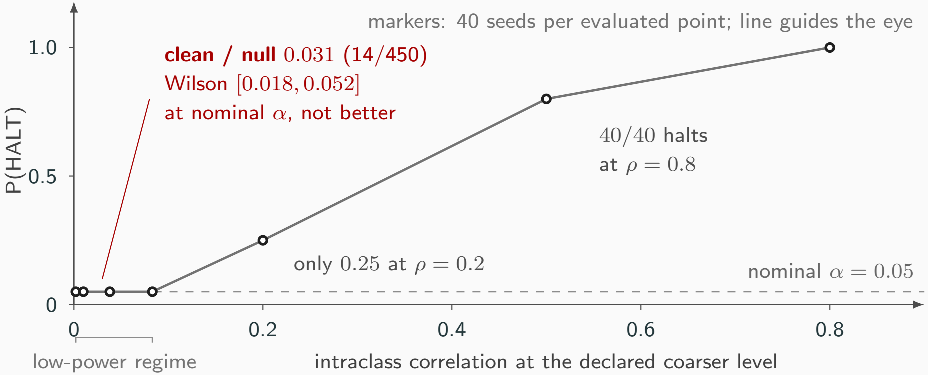
L4.7 — statistical-unit adequacy

Does dependence survive one declared level coarser? Null: exchangeability across those groups.

And it *abstains*: designs below the registered resolution floor return INDETERMINATE. That floor is an *attainability* property, not a power one.

One judgement generates both: neither obligation can be certified, so we accept checks that can only refuse.

Calibrating the gate before trusting it



14 halts in 450 clean evaluations. The interval, not the point, is the claim — and it contains α.

Does the unit problem appear in real orbital data?

source cohort

331 predicted visible passes

109 element-set records

11 objects

$\downarrow$ -59 passes lack a successor element set

primary in-track analysis

272 passes

in 90 successor-paired element sets

grouping	**verdict**
pass $\to$ element set 272 in 90	$\hat{\rho} = 0.501$ $p = 0.0025$ lower bound 0.376 **HALT**
element set $\to$ object 90 in 11	$\hat{\rho} = 0.284$ **HALT**

Both p at $B = 400$, the smallest the reference admits. Observable: the *in-track update increment* between consecutive fits sharing an observation arc — **not truth error**.

Exchangeability is rejected at *every* level tested here; the rule adjudicates the level it is *given*. On elevation the same rule returned $\hat{\rho} = 0.0$.

Can the frozen contract be applied outside this project?

A public spacecraft-telemetry LSTM artifact, frozen at a commit chosen *before* inspection. The detector `sha256` is unchanged throughout — equal to the pre-registration hash.

5 PASS	3 HALT	1 INDET	5 N/A	5 N/OBS

rule verdicts **applicability dispositions**

5 of 19 rules are recorded *not observable* rather than passed. Non-observability is never scored as compliance.

A *disposition* precedes any verdict: the obligation does not arise, or the artifact does not expose the inputs the rule needs. **3 HALT is a verdict on the recorded experiment, not on the published result.**

The contract runs unchanged on an artifact nobody here produced.

Does correcting a detected violation change the experiment?

	original		corrected
overlap	100.0% of val. support	→	0
L4.1	HALT	→	PASS
checkpoint	changed in 5 of 5 paired seeds		

Re-running one seed reproduces its checkpoint **bit for bit** — the intervention, not training noise.

The reported detection metric moved on three seeds, *in both directions*.

Not a null: at 5 paired seeds the smallest attainable two-sided p is $0.0625 > \alpha$.

$\Rightarrow$ **not estimable** at the available paired-run resolution — an evidentiary limit of *this study*, not a rule verdict and not a null result.

Model selection changed; the downstream endpoint is not estimable at this resolution.

What is actually new

Not the primitives.

ICC(1), permutation inference, metamorphic relations, hermetic build closure, mutation testing — all established, none proposed here.

Not a checklist.

The 19 rules are the satellite *instantiation*, not the contribution.

The method is the conversion

1. Identify which validity obligations are *relational*.
2. Choose the counterfactual execution or coarser level that makes each *falsifiable*.
3. Encode the invariant as a contract — and let an undecidable case return INDETERMINATE rather than pass.

A permutation reference is ordinarily an analysis step. Here it calibrates a *gate* — with an operating characteristic and an abstention state.

What this does not establish

- **No completeness claim.** L4.6 is bounded by the caller's variation set; the input nobody thought of is absent by construction.
- **No RF or packet-level result.** The object of study is the experiment, not the link.
- **No improvement in learned accuracy.** The contract refuses claims; it does not make models better.
- **Represented-fault regression coverage only.** Faults and rules largely share an origin, so curated coverage measures *reachability*, not fault-finding power.
- **One third-party artifact, one channel.** It partially breaks the loop. It does not close it.
- **PASS means not rejected**, never verified.

A validity method must apply the same evidence discipline to its own claims.

Chronological separation remains necessary.
It is not sufficient.

Some assumptions must become **executable**, **falsifiable**,

and allowed to **refuse** an unsupported claim.

Backup — why the abstention floor is not a power floor

Take the specific construction of *three coarser groups of two units*: it admits $6!/(2!^3 3!) = 15$ distinct assignments, so the smallest attainable p is $1/15 = 0.067 > \alpha$. That design cannot reject at the nominal level at any effect size.

Four groups of two admit 105 assignments, whose exhaustive reference reaches 0.0095. The general rule is stated at the rule level, not by extrapolating $1/15$: designs below the registered resolution floor return INDETERMINATE.

So INDETERMINATE is a statement about the *design's resolution*, not about the strength of the effect. Collapsing it into PASS is how a design that could not have rejected comes to look like one that did not.

Backup — reproduction and provenance

- `make gate` regenerates the fault matrix and L4.7's operating characteristic, and *refuses to build* if a headline number disagrees with the artifact at the sentence claiming it. The manuscript contains 64 artifact-bound claim sites; every numerical value shown in this deck is generated from that same summary artifact.
- The two external studies are hash-matched rather than regenerated: their inputs do not ship.
- `make external-consequence-verify` checks the committed third-party results against the frozen commit, the frozen detector hash and the recorded data hashes — 16 checks, no training.
- The telemetry arrays came from a *checksum-verified mirror* after the upstream endpoint became unavailable; per-file hashes matched two independently published checksum sources. That is mirror concordance, *not* verification by the original publisher.